Lecture 22 2025-11-25

4.5 Romberg Integration

- Combine Richardson extrapolation with composite trapezoidal rule

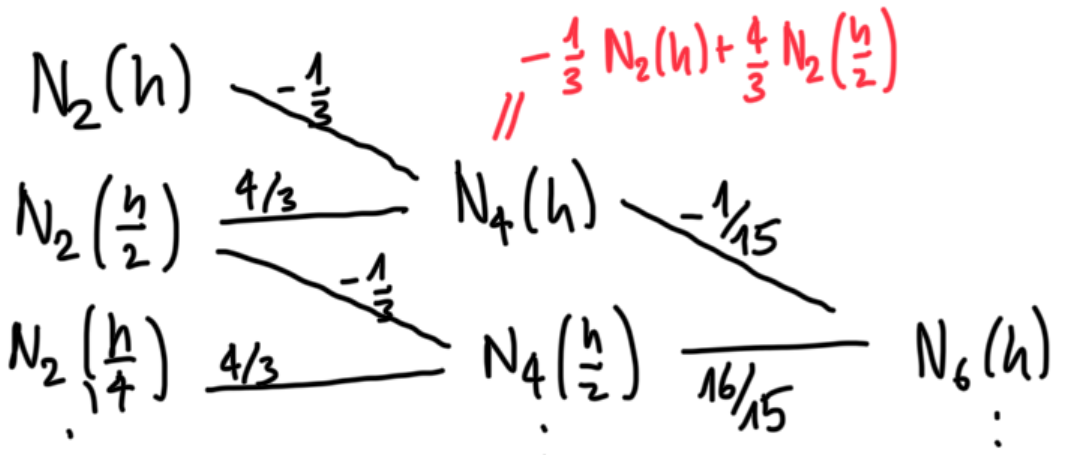
We saw that $\int_a^b f(x)dx = \underbrace{T(h)}_{\text{composite trapezoidal rule}} - \frac{b-a}{12} f''(\mu) h^2$ for some $\mu \in (a,b)$

One can also show that $\int_a^b f(x)dx = T(h) + K_1 h^2 + K_2 h^4 + K_3 h^6 + \dots$

where each K_i is a constant depending only on $f^{(2i-1)}(a)$ & $f^{(2i-1)}(b)$

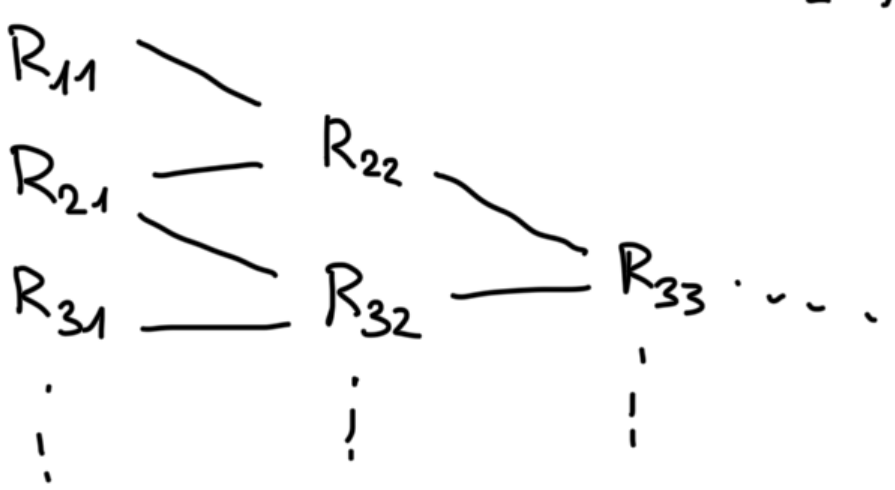
This looks perfect for Richardson extrapolation which is called Romberg Integration in this setting.

Now $N_2(h) = T(h)$, $N_{p+2}(h) = \frac{2^p N_p(\frac{h}{2}) - N_p(h)}{2^p - 1}$ $p = 2,4,6,8,\dots$

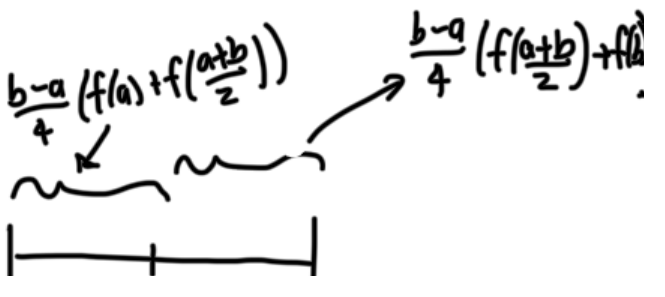


Romberg table Set $h = b-a$ ($h = \frac{b-a}{n}$ for composite trapezoidal rule)

$$R_{ij} = \begin{cases} 0 & i<j \\ N_{2^{i-j}}(\frac{h}{2^{i-j}}) & i \geq j \end{cases}$$



$$R_{ij} = \frac{4^{j-1}}{4^{j-1}-1} R_{i,j-1} - \frac{1}{4^{j-1}-1} R_{i-1,j-1}$$



Turns out $R_{22} = \frac{4}{3} R_{21} - \frac{1}{3} R_{11}$



$$= \frac{4}{3} \frac{b-a}{4} \left(f(a) + f(b) + 2 f\left(\frac{a+b}{2}\right) \right) - \frac{1}{3} \frac{b-a}{2} \left(f(a) + f(b) \right)$$

$$= \frac{b-a}{3} \left(\frac{f(a)}{2} + \frac{f(b)}{2} + 2 f\left(\frac{a+b}{2}\right) \right)$$ 'Simpson's Rule.'

R_{33} is also a Newton Cotes formula.

When $r \geq 4$, Romberg R_{rr} is more stable than Newton–Cotes with same number of degrees of freedom.

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5. Initial value problems for ordinary differential equations (ODEs)

Goal: Approximation of solution $y(t)$ of a problem of the form

$$\text{(ODE)} \quad \frac{dy(t)}{dt} = f(t, y(t)) \quad \text{for } a \leq t \leq b$$

subject to initial condition $y(a) = \alpha$

y can be scalar- or vector-valued, i.e.

$$y : [a,b] \to \mathbb{R} \quad \text{or} \quad \begin{cases} y(t) = (y_1(t), y_2(t), \dots, y_n(t)) : [a,b] \to \mathbb{R}^n \\ \text{and } f = (f_1, \dots, f_n) = (f_1(t, y_1, \dots, y_n), \dots, f_n(t, y_1, \dots, y_n)) \\ y(a) = (\alpha_1, \dots, \alpha_n) \end{cases}$$

Example: Population growth

$$\frac{dy}{dt} = \kappa y$$

y: population size

κ: growth rate

(unrestricted growth)

Rate of change in population size depends on current size (e.g. bacteria)

Solution: $y(t) = \alpha e^{\kappa(t-a)}$ (if $y(a) = \alpha$)

This is a 1st order ODE. We will focus on the approximation of solutions of these. Higher order ODEs can be cast into 1st order systems as

follows:

Say $y^{(n)} = f(t, y, y', y'', y''', \dots, y^{(n-1)})$ (*)

$y(a) = \alpha_1,\ y'(a) = \alpha_2, \dots, y^{(n-1)}(a) = \alpha_{n-1}$

Define $z_1 = y,\ z_2 = y', \dots, z_n = y^{(n-1)}$, $z = (z_1, \dots, z_n)$

$z(a) = (z_1(a), \dots, z_n(a)) = (\alpha_1, \dots, \alpha_n)$

(*) $\Rightarrow$
$$\begin{cases} z_n' = f(t, z_1, \dots, z_n) \\ z_1' = z_2 \\ z_2' = z_3 \\ \vdots \\ z_{n-1}' = z_n \end{cases}$$

$$\begin{cases} z' = F(t, z) \\ \\ F(t,z) = (z_2, z_3, \dots, z_n, f(t,z)) \end{cases}$$

Existence & uniqueness of solutions of (ODE)

Def 5.1 A function $f(t,y)$ satisfies a Lipschitz condition in the variable y on a set D if there exists $L > 0$ with

$|f(t,y) - f(t,\tilde{y})| \leq L\,|y - \tilde{y}| \quad \forall\ (t,y), (t,\tilde{y}) \in D$

L is called Lipschitz constant for f

Picard-Lindelöf theorem (Thm 5.4 in textbook) (Existence & uniqueness of solutions of (ODE))

Let $D = \{(t,y) \mid a \leq t \leq b,\ y \in \mathbb{R}^n\}$ $n \in \mathbb{N}$

and assume f is continuous on D. If f is Lipschitz continuous in the variable y, then (ODE) has a unique solution $y(t)$ on $a \leq t \leq b$.

Wellposedness (Hadamard Notion)

(ODE) is said to be wellposed if

(i) A unique solution $y(t)$ to the problem exists

(ii) There exists constants $\varepsilon_0 > 0$ and $k > 0$ s.t. $\forall\ \varepsilon \in (0, \varepsilon_0)$

& $\delta(t)$ continuous with $|\delta(t)| < \varepsilon \quad \forall t \in [a,b]$ and $|\delta_0| < \varepsilon$

then $\left.\begin{aligned} \frac{dz}{dt} &= f(t,z) + \delta(t) \qquad a \le t \le b \\ z(a) &= \alpha + \delta_0 \end{aligned}\right\}$ (*) → perturbed problem associated with (ODE)

has a unique solution $z(t)$ that satisfies

$$|z(t) - y(t)| < k\varepsilon \qquad \forall t \in [a,b]$$

"Perturb f and α a little, still get a unique solution that is not very different from solution of original problem"

Thm 5.6 D as before, f continuous in t and y, f Lipschitz with respect to y, then (ODE) is well-posed.

Example: $\frac{dy}{dt} = \mathcal{K}y - t^2 + 1 := f(t,y)$ continuous wrt t & y

$$|f(t,y) - f(t,\tilde{y})| = |\mathcal{K}y - t^2 + 1 - (\mathcal{K}\tilde{y} - t^2 + 1)|$$
$$= |\mathcal{K}(y - \tilde{y})| = |\mathcal{K}|\,|y - \tilde{y}|$$

so f is Lipschitz with Lipschitz constant $L = |\mathcal{K}|$

⇒ this ODE is well-posed.

5.2. Euler's Method

Simple way to discretize (ODE) (based on Taylor's theorem)

We will find approximations of $y(t)$ at various values in $[a,b]$ called mesh points. Once approximation is found at mesh points, an approximate solution can be found at other points in the interval via interpolation.

Assume mesh points are equally spaced throughout $[a,b]$, i.e. pick $N \in \mathbb{N}$ & set $h = \frac{b-a}{N}$. Then define

$$t_i = a + ih \quad , i = 0, 1, \ldots, N$$

$h = t_{i+1} - t_i$ is called step size or time step size

Assume $y(t) \in C^2([a,b])$. Then by Taylor's theorem

$$y(t_{i+1}) = y(t_i) + \underbrace{(t_{i+1}-t_i)}_{h}\,\underbrace{y'(t_i)}_{f(t_i,y(t_i))} + \frac{(t_{i+1}-t_i)}{2}\,y''(\xi_i) \qquad \text{where } \xi_i \in (t_i, t_{i+1})$$

$$= y(t_i) + h f(t_i, y(t_i)) + \underbrace{\frac{h^2}{2}\,y''(\xi_i)}_{O(h^2) \text{ if } h \text{ small}}$$

Now the forward/explicit Euler method is obtained by removing the remainder term which is heuristically small

So $y_i \approx y(t_i)$ (book $w_i \hat{=} y_i$)

(FE)
$$\begin{cases} y_0 = \alpha \\ y_{i+1} = y_i + h f(t_i, y_i) \qquad i = 0, 1, 2, \ldots \end{cases}$$